Adversarially-Robust TD Learning with Markovian Data: Finite-Time Rates and Fundamental Limits

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Problem and method. We study policy evaluation with linear value-function approximation from a single Markov trajectory. An adversary can replace each observed reward by an arbitrary value with probability ϵ, while clean rewards need only have bounded second moments. The goal is to learn the TD fixed point despite temporal dependence and corruptions that can drive vanilla TD toward an arbitrary parameter. **Robust-TD** replaces the instantaneous reward-feature vector by a robust estimate of its stationary expectation. It applies RUMEM separately to each coordinate of the reward-feature history: subsampling reduces dependence, and median-of-means aggregation limits the effect of outliers. After an initial burn-in, extreme estimates are reset to zero using a dynamic threshold; the resulting estimate enters the usual TD matrix update.

Assumptions and notation. The policy induces an ergodic Markov chain, initialized at stationarity. Features satisfy $\|\phi(s)\|_2 \le 1$ and have full column rank; their dimension is d. Mean rewards obey $|R_\mu(s)| \le \bar{r}$, reward variance is at most ρ^2, and each reward is arbitrarily corrupted with probability ϵ. Let $D = \mathrm{diag}(\pi)$, $\omega = \lambda_{\min}(\Phi^\top D \Phi) > 0$, and

$$\sigma_1 := \max\{1, \bar{r}, \rho\}, \qquad \sigma := \max\{\|\theta^\star\|_2, \|\theta_0\|_2, \sigma_1\}, \qquad G := \frac{d}{\omega^2(1-\gamma)^2}.$$

Robust TD update. Coordinate-wise RUMEM estimates $\bar{b} = \mathbb{E}_\pi[\phi(s) r]$ from $\{[\phi(s_k)]_i \widetilde{r}_k\}_{k \le t}$. After burn-in, threshold $\widehat{b}_t$ and update

$$\theta_{t+1} = \theta_t + \alpha(A_t \theta_t + \widehat{b}_t), \qquad A_t = \phi(s_t)\big(\gamma\phi(s_{t+1}) - \phi(s_t)\big)^\top, \qquad \theta^\star = -\bar{A}^{-1}\bar{b}.$$

1. Mean-square finite-time guarantee. Set $\alpha = 4\log T/[\omega(1-\gamma)T]$ and burn-in $\bar{T} = \lceil c_1 \tau_{\mathrm{mix}} \log^2(dT)\rceil$. Let $\tau' = \tau'_{\mathrm{mix}}(\alpha)$ be the mean-operator mixing time and $\bar{\tau}_{\mathrm{mix}} := \max\{\tau', \tau_{\mathrm{mix}}\}$. For $T \ge \max\{\bar{T} + \tau', c_2 \tau' \log T/[\omega^2(1-\gamma)^2]\}$,

$$\mathbb{E}\|\theta_T - \theta^\star\|_2^2 \le \widetilde{O}\left(\frac{\bar{\tau}_{\mathrm{mix}} \sigma^2 G}{T}\right) + O\big(\epsilon \sigma_1^2 G\big). \tag{1}$$

The statistical term decays as $\widetilde{O}(1/T)$; the squared-error corruption term is $O(\epsilon)$.

2. Fundamental policy-evaluation limit. Even in an i.i.d. tabular observation model, there is a universal $c > 0$ such that

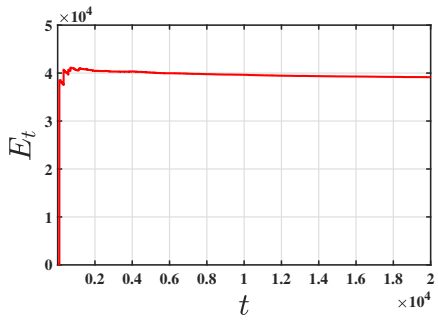
$$\inf_{\widehat{V}_T} \sup_{\mathcal{M}(\epsilon,\rho)} \mathbb{P}\left(\|\widehat{V}_T - V\|_2 \ge \frac{c\rho\sqrt{\epsilon}}{1-\gamma}\right) \ge \frac{1}{2}. \tag{2}$$

Thus an ϵ-order squared-error penalty cannot generally be eliminated; the lower bound concerns value estimation in the tabular subclass.

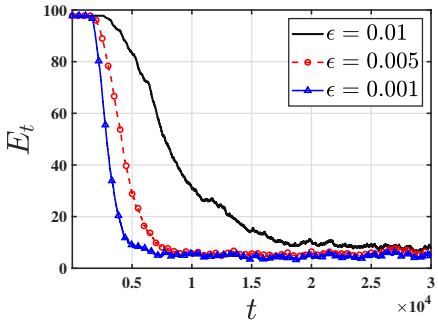
3. Robust mean estimation with dependent, corrupted data. For n observations of a scalar Markov process with bounded clean mean function $|f| \le \psi$ and noise variance at most ρ^2, RUMEM uses gap $\tau = \lceil \log_2(6n/\delta)\tau_{\mathrm{mix}} \rceil$ and suitable MoM buckets. Under its sample-size conditions, with probability at least $1-\delta$,

$$|\widehat{\mu} - \bar{f}| \le C \max\{\psi, \rho\} \left(\sqrt{\epsilon} + \sqrt{\frac{\tau}{n}\log(n/\delta)}\right). \tag{3}$$

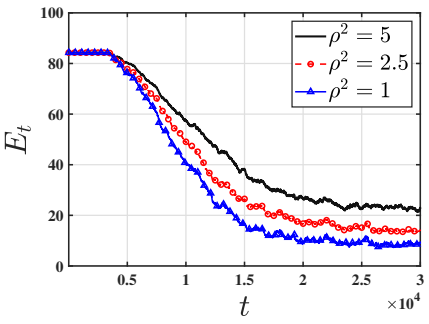
Numerical results. 100 states, $d = 10$, $\gamma = 0.5$, and 10 independent trials; the attacker injects $100/\epsilon$. (a) TD(0) uses $\alpha_t = 1/t$ and $\epsilon = 0.001$. (b,c) Robust-TD uses $\alpha = 0.1$. Plots show mean-square error $E_t = \|\theta_t - \theta^\star\|_2^2$.



(a) TD(0), $\alpha_t = 1/t$.



(b) Robust-TD: varying ϵ.



(c) Robust-TD: varying ρ^2.